عدسة محدبة

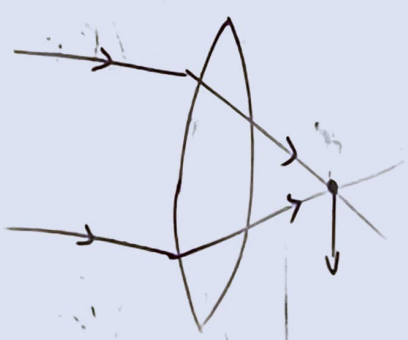
Convex lens

Aim: Determination of

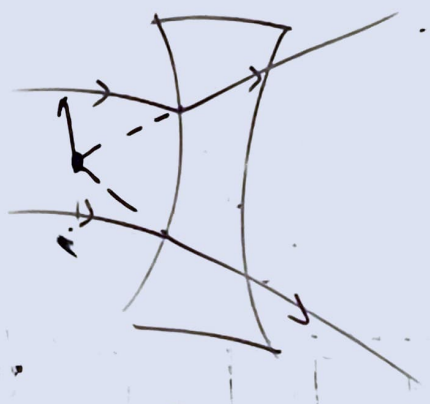
① Focal length (f)

② lens power (F)

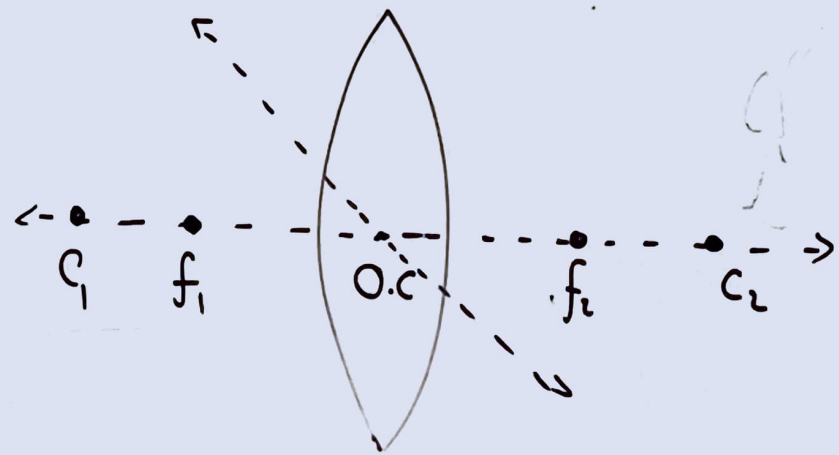
Theory :-

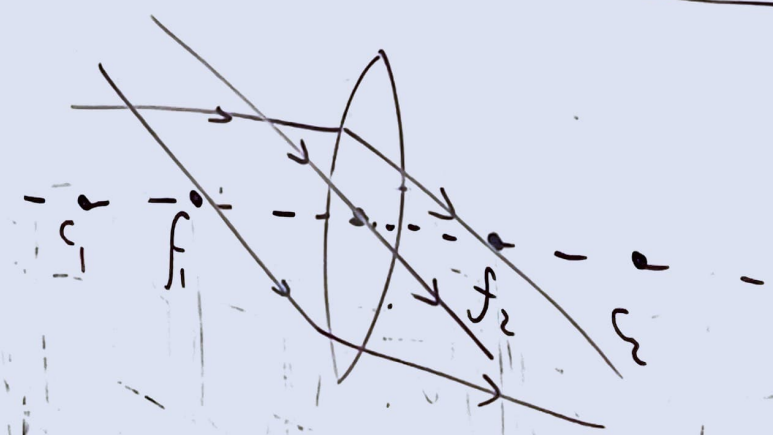


Convex



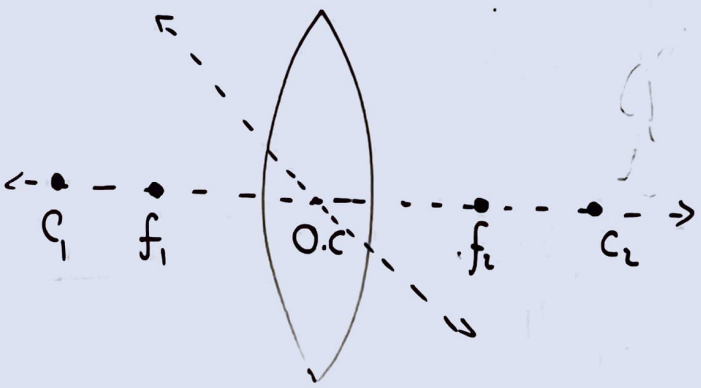
Concave

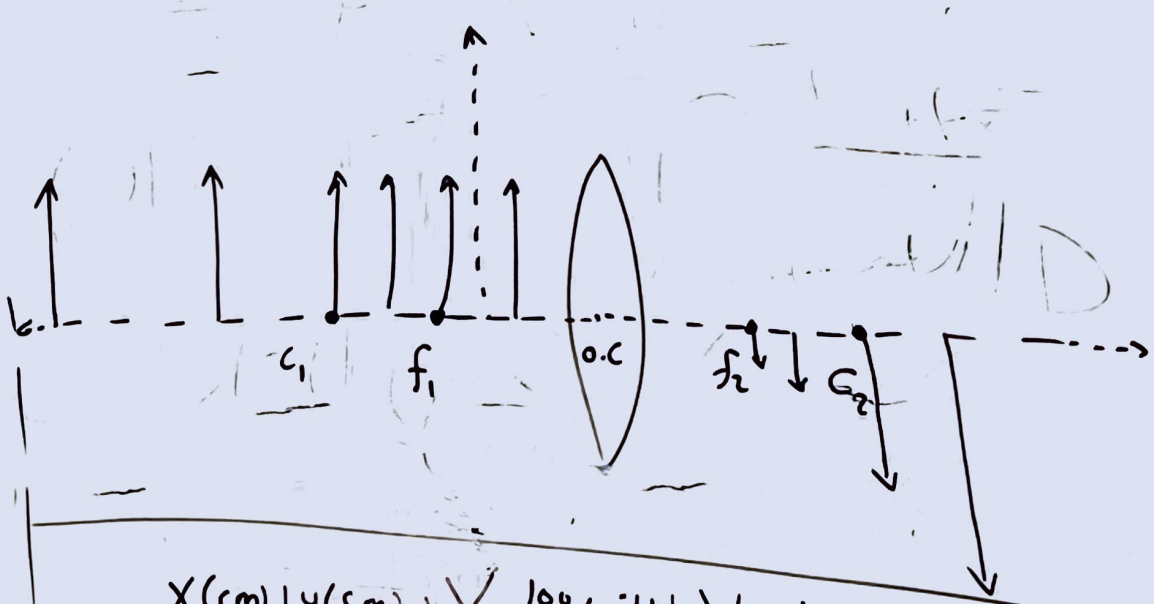


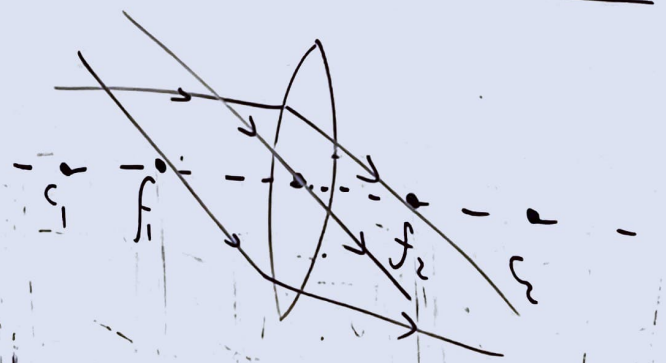


عدسة محدبة

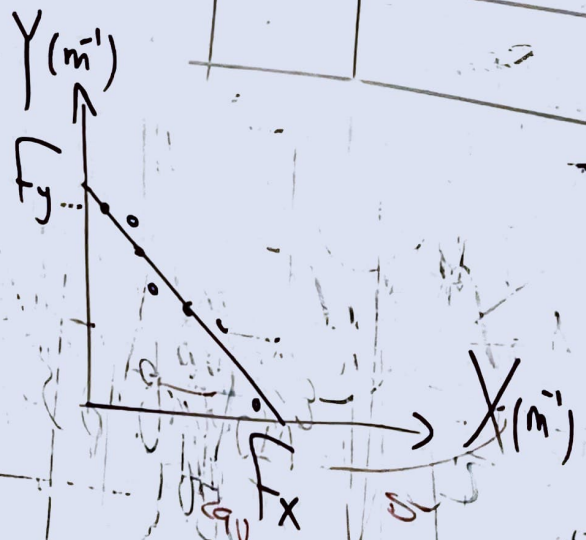
Convex lens







X(cm)	y(cm)	$X = \frac{100}{x}(m^{-1})$	$Y = \frac{100}{y}(m^{-1})$



$$F = \frac{f_x + f_y}{2} \quad \ldots\ldots (m^{-1})$$

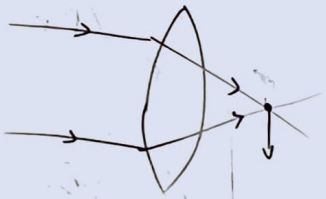
$$f = \frac{1}{F} \quad \ldots\ldots m \quad \ldots\ldots cm$$

عدسة محدبة

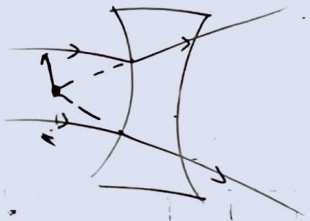
Convex lens

Aim: Determination of
1. Focal length (f)
2. lens power (F)

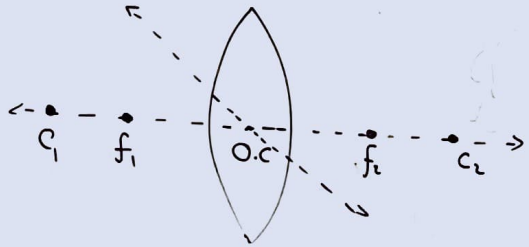
Theory :-

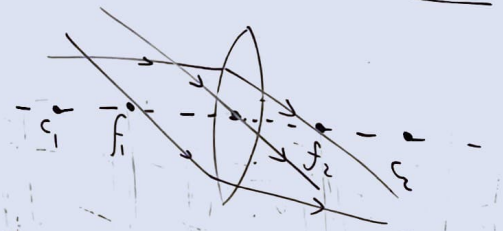


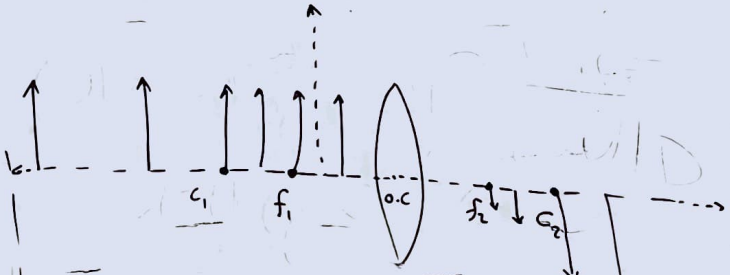
Convex



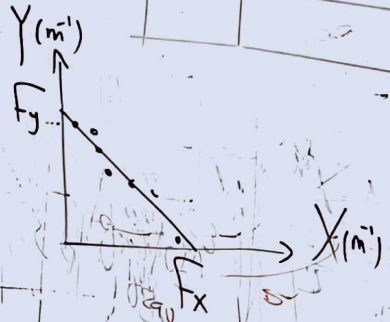
Concave





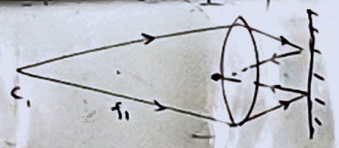


X (cm)	y (cm)	$X = \frac{100}{x}$ (m^{-1})	$Y = \frac{100}{y}$ (m^{-1})



$$F = \frac{F_x + F_y}{2} = \ldots\ldots m^{-1}$$

$$f = \frac{1}{F} = \ldots\ldots m = \ldots\ldots cm$$



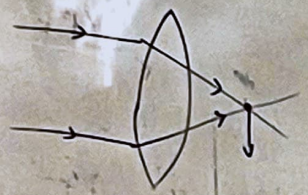
عدسة محدبة

Convex lens

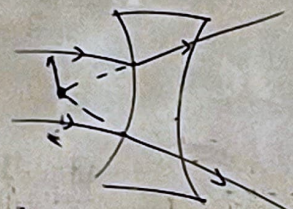
Aim: Determination of

1. Focal length (f)
2. lens power (F)

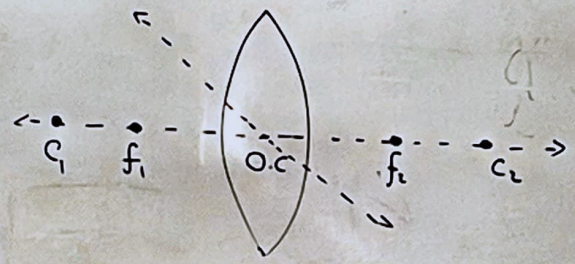
Theory:-

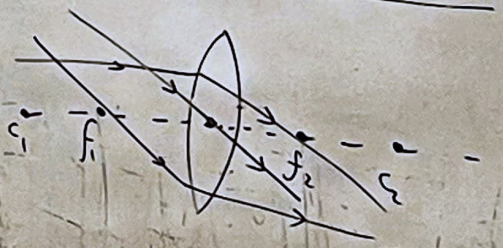


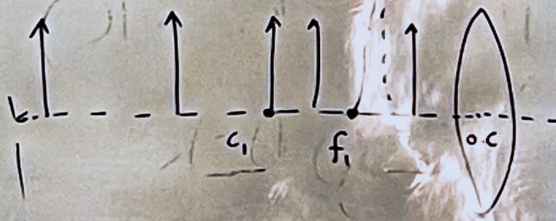
Convex



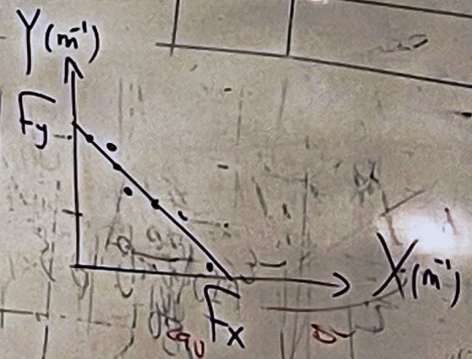
Concave







X(cm)	y(cm)	$X = \frac{100}{x}$ (m^{-1})



Convex Lens

Aim ① Determin The Focal length of lens (f)

② // The lens Power at different distance (F)

Law $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

$F = \frac{100}{f}$

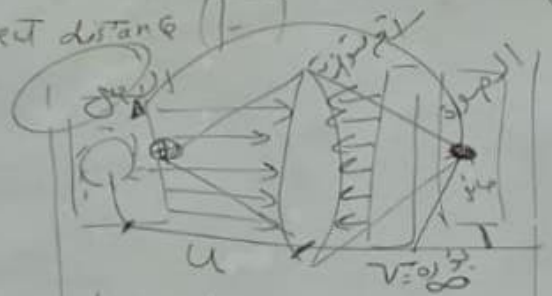
Theory ① Types of lenses

② Focal length

③ Method of measure

Ⓐ Identical Method

Ⓑ General or graphyical Method



$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

$\frac{1}{f} = \frac{1}{\infty} + \frac{1}{v}$

$f = v$

① الطريقة الاولى

$\frac{1}{f}$ $u + \frac{1}{\infty}$

$f = u = 33$ cm

$F = \frac{100}{33}$ (cm^{-1})

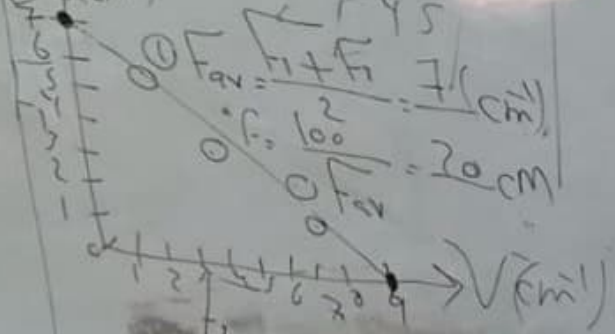
②

$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$ $*100$

$\frac{100}{f} = \frac{100}{u} + \frac{100}{v}$

$F = U + V$

U	V	$U = \frac{100}{u}$	$V = \frac{100}{v}$
30	36		
35	29		
40	25		



① $F_{av} = \frac{F_1 + F_2}{2} = 7$ (cm^{-1})

$f = \frac{100}{F_{av}} = 30$ cm